Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *NLLB* | ***Context:*** *Amhara Region Agricultural and Microfinance Translation Cohort*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's input form is well-aligned with the deployment at the modality and script levels. All inputs are text-only encoded via BCP 47 with ISO 639-3 + ISO 15924 script subtags [Q79], the Ethiopic (Ge'ez) script is explicitly supported with a dedicated 8k SentencePiece vocabulary [Q382], and the deployer confirmed text-only prose input is acceptable [elicitation A4]. Average sentence length (~21 words) [Q133] reflects general-domain prose. The deployment is HIGH on text-only operation, and the benchmark matches. Residual concerns are: (a) sentence-level evaluation architecture does not assess document-level structures (numbered clauses, conditional logic), and (b) Flores-200 sentences are general-domain prose, whereas administrative source documents have higher punctuation density and clause structure — though deployer accepted prose output [elicitation A4]. IF was assigned LOWER priority by elicitation, consistent with this score.

ID	Evidence
Q79	“We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script.” (p.17)
Q133	“On average, sentences are approximately 21 words long.” (p.22)
Q382	“We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script.” (p.45)

Strengths

- Dedicated 8,000-token Ge'ez SentencePiece vocabulary trained jointly for Amharic and Tigrinya [Q382] addresses a key low-resource representation issue.
- BCP 47 / ISO 639-3 / ISO 15924 standardization [Q79] gives unambiguous language and script identification, important since Amharic uses Ethiopic exclusively.
- Combined supervised cosine-loss + unsupervised MLM training procedure [Q373, Q374] leverages monolingual Amharic data, which is more plentiful than bitext.

ID	Evidence
Q79	“We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script.” (p.17)
Q373	“We also deployed a new training procedure which combines supervised training, i.e. minimizing the cosine loss between the teacher and student embedding, and unsupervised masked LM training (see left part of Figure 12).” (p.44)
Q374	“This enabled us to benefit from monolingual data during encoder training.” (p.44)
Q382	“We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script.” (p.45)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target

IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions broadly match: text-only, Ethiopic script, sentence-level prose [Q79, Q133]. Source documents in deployment are also text in Ethiopic script. Sentence length distribution may differ (administrative documents have longer, clause-heavy sentences) but benchmark sentences average ~21 words [Q133] which is in a reasonable range.

IF-2: Regional infrastructure can support text input; the deployment is text-only. However, web search [WEB-7, WEB-8] documents recurring mobile internet shutdowns in Amhara (August 2023–July 2024 and April–May 2023) which could disrupt connected pipelines, but this is an operational risk rather than a benchmark-form mismatch.

IF-3: Domain-specific form differences: administrative source documents contain numbered clauses, tables, and cross-references absent from Flores-200 sentences [Q95, Q133]. The benchmark does not evaluate document-level coherence. The deployer accepted prose output [elicitation A4], reducing the severity of this gap.

IF-4: Form mismatch is minor at the script/modality level. The principal residual gap is the absence of document-level evaluation, which is more accurately a scoring/output issue than an input-form issue.

ID	Evidence
Q79	“We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script.” (p.17)
Q95	“As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark.” (p.19)
Q133	“On average, sentences are approximately 21 words long.” (p.22)
WEB-7	Freedom House 2024: documented Amhara mobile internet shutdowns Aug 2023–July 2024 (operational risk, not benchmark form mismatch) (https://freedomhouse.org/country/ethiopia/freedom-net/2024)
WEB-8	Freedom House 2023: prior April–May 2023 connectivity blackouts in Amhara (https://freedomhouse.org/country/ethiopia/freedom-net/2023)

Information Gaps

- Sentence-length distribution comparison between Flores-200 and actual administrative Amharic documents is not documented.